BHUVANESH ARUN C T

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Education

B.E -Electrical and Electronics Engineering **CGPA: 8.63**
Chennai Institute of Technology **2023-2027**

Research Activity

Conducted research on *"A comprehensive review of power quality challenges arising from high-penetration electric vehicle integration and strategies for resilient grid adaptation,"* which has been *submitted to the journal Energy Exploration & Exploitation* for publication consideration.

Internship Experience

Schneider Electric – Production Intern ***Dec 2025 - Present | Bangalore, Karnataka***

- Currently pursuing an internship at Schneider Electric in the Production Department, where I developed dashboards and operational tools to improve plant efficiency. Contributed to production optimization by supporting data-driven decisions and coordinating with cross-functional teams.

PICABORD Technologies Pvt. Ltd. - Hardware Design Engineer ***April 2025 -Dec 2025 | Chennai, Tamil Nadu***

- Created the basic layout of a single board computer, including schematic development and PCB design. Gained hands-on experience in hardware integration, component selection, and system-level embedded design.

Meera Laser Solutions Pvt. Ltd - Project Engineer Intern ***Aug 2025 - Sep 2025 | Chennai, Tamil Nadu***

- Worked on the setup of a laser welding head on a KUKA robotic arm, ensuring seamless execution by coordinating with partner companies and managing resource arrangement, equipment arrival, and vendor alignment, while gaining exposure to project coordination, industrial automation, and collaborative execution in a manufacturing environment.

Renom Wind Energy Services Pvt Ltd - Engineering Trainee ***Nov 2024 – Dec 2024 | Tirunelveli, Tamil Nadu***

- Completed a 6-week internship on wind turbine operation and maintenance, gaining hands-on experience with Enercon E40 models. Assisted in inspections and troubleshooting, enhancing knowledge of electricity generation and renewable energy systems.

Vels Wind Power - Lab Engineer Intern ***May 2024 - June 2024 | Kanyakumari, Tamil Nadu***

- Completed a 45-day internship as a lab technician engineer, gaining hands-on experience in applying theoretical knowledge. Assembled and tested a 12V battery charger circuit PCB while developing skills in circuit troubleshooting, error detection, and rectification.

Project

Custom Arduino Nano PCB Design using KiCad Designer

Technologies: KiCad, Gerber generation

- Designed and developed a custom Arduino Nano-compatible PCB using KiCad, covering schematic design, optimized PCB routing, 3D visualization, and Gerber file generation, enabling seamless hardware prototyping and manufacturing.

Power System Load Flow Simulator (Full-Stack Web Application)

Technologies: React, Node.js, Express, React Flow, Zustand, Vitest, GitHub Pages, Render.

- Designed and developed an interactive load flow simulator with drag-and-drop network modeling, Y-bus generation, Gauss-Seidel analysis, convergence visualization, and PDF export; built with prompt-driven development and deployed with frontend on GitHub Pages and backend API on Render.

Smart Protection System for Electrical Appliances using IoT and Machine Learning

Technologies: ESP32, DS18B20 sensor, Current/Voltage sensor, SSR/Relay, Web Dashboard (HTTP), Firebase, Machine Learning

- Developed an IoT-enabled protection system for electrical appliances using ESP32 and sensors for real-time temperature, current, and voltage monitoring. Integrated automatic power disconnection, remote monitoring through a web dashboard, alert notifications, and machine learning-based analysis to detect overheating risks and prevent electrical hazards.

Cost-Affordable IoT-Based Nutrient Analysis Kit for Fermented/Liquid Organic Manure

Technologies: ESP32, TCS3200 colour sensor, reagent-based colorimetry, web dashboard, calibration curve

- Created a low-cost IoT system to quantify nutrient levels in Fermented/Liquid Organic Manure (F/L OM) based on a problem statement from Smart India Hackathon (SIH). The device employs a TCS3200 color sensor and ESP32 to detect reagent-induced color changes to arrive at NPK levels. A web interface offers real-time monitoring and suggestions, facilitating effective manure applications for farmers.

Technical Skills

Hardware & PCB: Altium Designer, KiCad, PCB routing, Gerber generation, PCB assembly, LTspice, Fritzing | **Embedded & IoT:** ESP32, Arduino , Sensor integration, UART/I2C/SPI, REST/HTTP dashboards | **Programming & Simulation:** C (Intermediate), C++ (Basic), Java (Basic), MATLAB/Simulink | **Automation & Robotics Tools:** RoboDK, KUKA robot experience (KR 16, LBR Issy), Fusion 360 | **Platforms & OS:** Linux, Web dashboards (basic HTML/CSS/JS), Git/GitHub

Certification

Universal Robotics | Introduction Energy Literacy Training | Cisco Cybersecurity Essentials & Introduction | Altium PCB Design | NPTEL Introduction to IoT | NPTEL Digital Circuits | AICTE Embedded System Developer Virtual Internship

Achievements

- Winner in Roborace event in Takshashila 2025 held at Chennai Institute of Technology.
- Finalist in Nexathon 2024 conducted at Chennai Institute of Technology
- Finalist in Autorace (RC racing) event in Kurukshektra 2025 held at College of Engineering, Guindy.
- Finalist in Deepsprint Hackathon (Top 30 out of 800) conducted by CIT Innovation Labs at Innovest 2025.

Extra-Curricular Activities

- Attached to KUKA's Centre of Excellence. Experienced in handling KUKA KR 16 and KUKA LBR Issy 11 robots
- Organized "Automation and Robotics Workshop" by KUKA Training Centre in Innovest 2024.
- Organized "Impulse 2025 (Symposium)" as Joint Secretary and Designer, using Adobe Photoshop and other design tools.